## MERN Stack Hands-on Task:

You are required to build a Collaborative Document Annotation System using the MERN stack. This problem is designed to test your end-to-end skills.

Requirements:

• Document Upload & Storage: A user can upload text or PDF files.

• Annotation Feature: On the frontend, users should be able to select a portion of text in the

document and add annotations/comments. Each annotation should be linked to the exact text range, user who made the annotation, and timestamp.

• Real-Time Collaboration: Multiple users can annotate the same document at the same time. Annotations should appear in real time to all connected users.

• Handle large documents efficiently. Annotations should remain accurate even if multiple users annotate overlapping text ranges. Prevent duplicate annotations for the same range by the same user.

• Optimizations Expected:

Efficient MongoDB schema design.

Backend APIs with proper good performance.

UI should not lag when loading documents with 1000+ annotations.

Proper error handling.

Deliverables:

• A working MERN app (doesn't need to be production-grade but should show solid architecture).

• Backend: Node.js + Express APIs with MongoDB schema.

• Frontend: React app with annotation UI (highlight text, add comments, show comments in a side panel).

• Real-time sync.

• Documentation: Short README explaining design choices (schema, performance optimizations, edge case handling).

• Share the link to a public GitHub repository with your source code.

• Deploy the application (any free platform) and share the live URL.

Please ensure your solution demonstrates both functional correctness and thoughtful technical design.

Pay attention to performance, user experience, and code readability.